

CRIME TRENDS IN LOS ANGELES (2024)

Team 5 – Harshith · Muhannad · Ebrima

CRIME SCENE DO NOT CROSS



Introduction

Goal

- Our goal in this project was to analyze 2024 Los Angeles crime data to understand trends, identify meaningful patterns, and detect statistically significant changes in crime activity across the city.
- Apply statistical tests and modeling to detect real trends and support data-driven insights.

Dataset:

- Over 134,000 crime records and 28 variables from LAPD Open Data Portal including descriptions, locations, victim info, and dates.
- Key fields used: crime type, victim age/sex, time of occurrence, location coordinates, and report timing.

SMART

QUESTION

- Which Areas had the highest crime concentration in 2024
- Which crime types showed the sharpest increases or decreases in 2024?
- Do crime occurrences in Los Angeles differ between Weekday and Weekends?

Data Cleaning

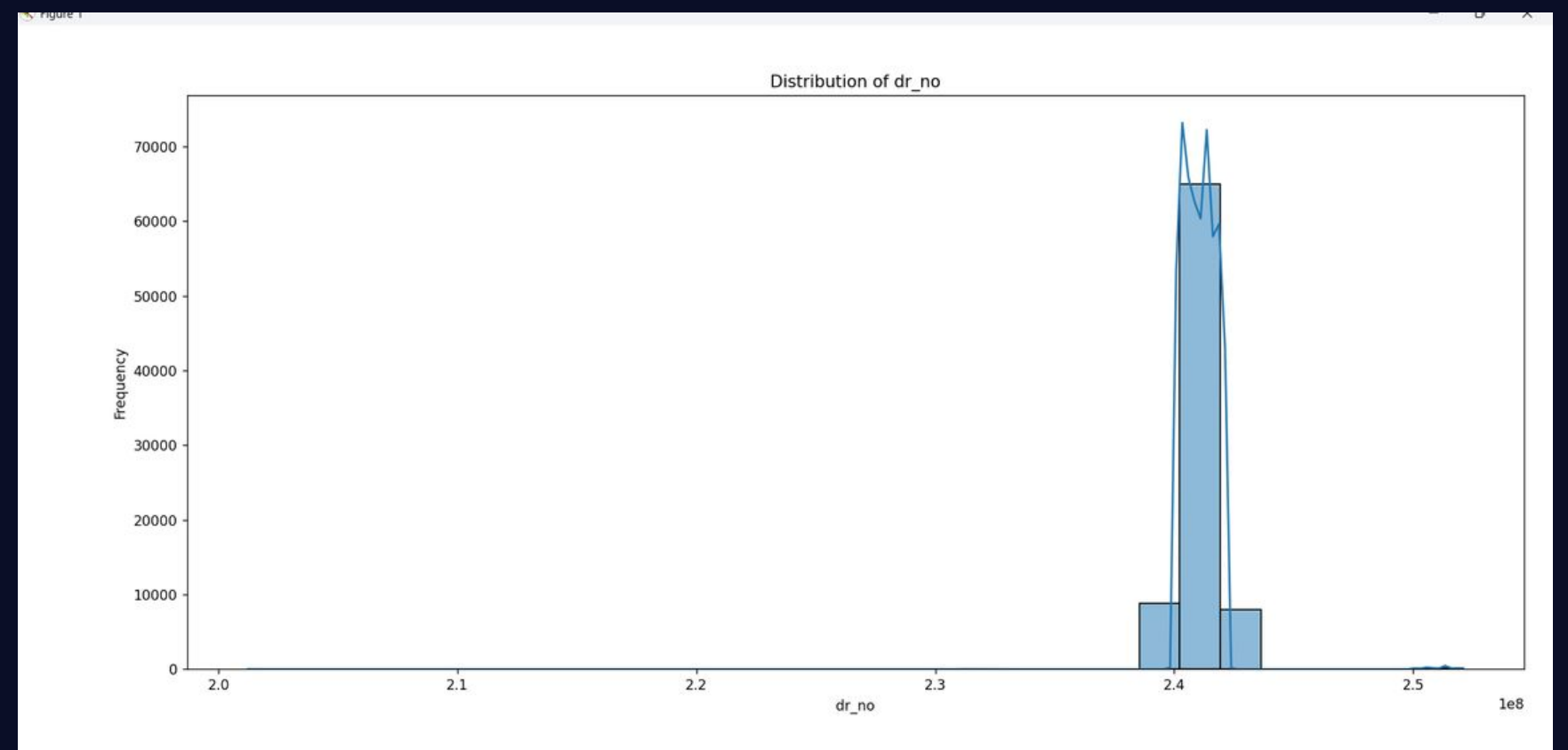
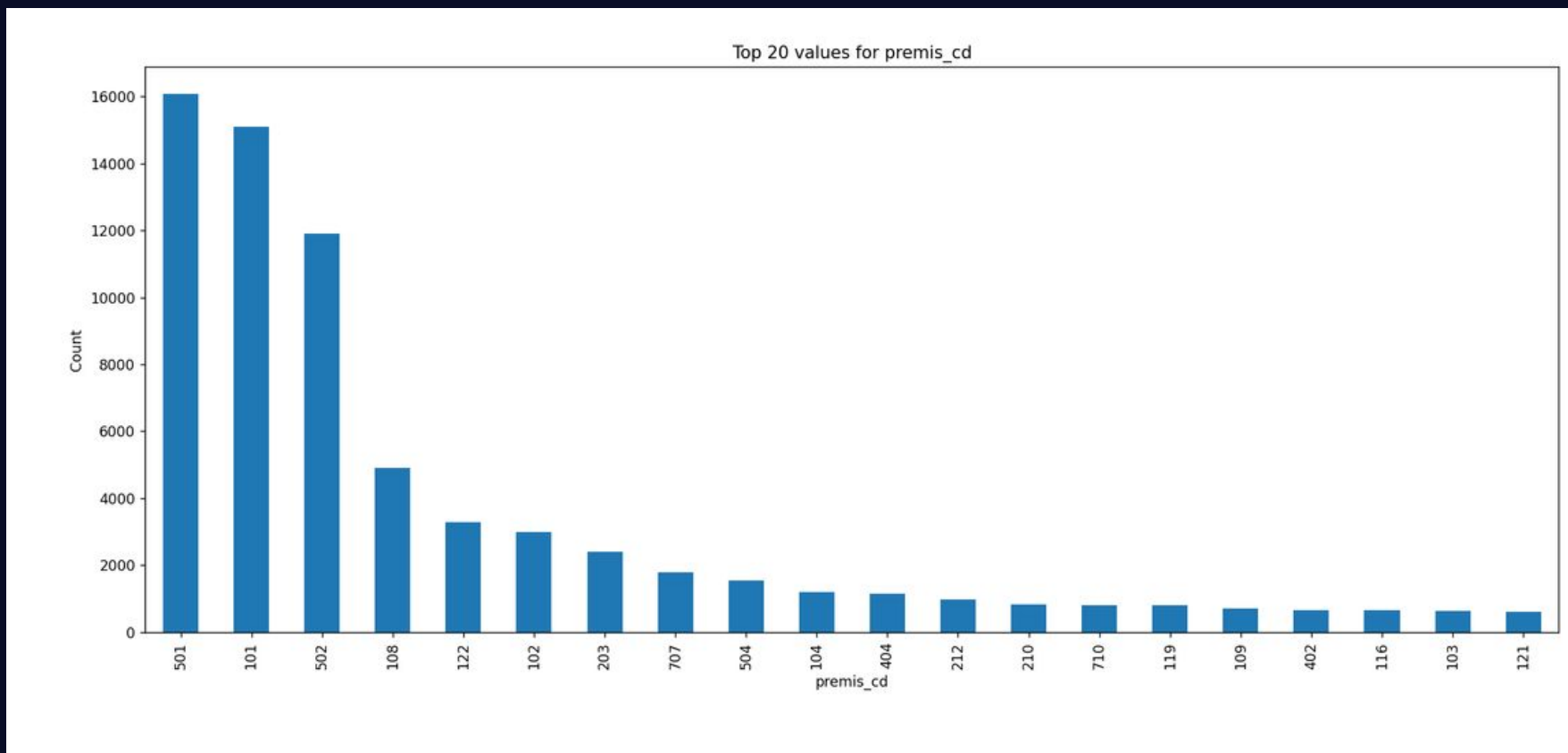
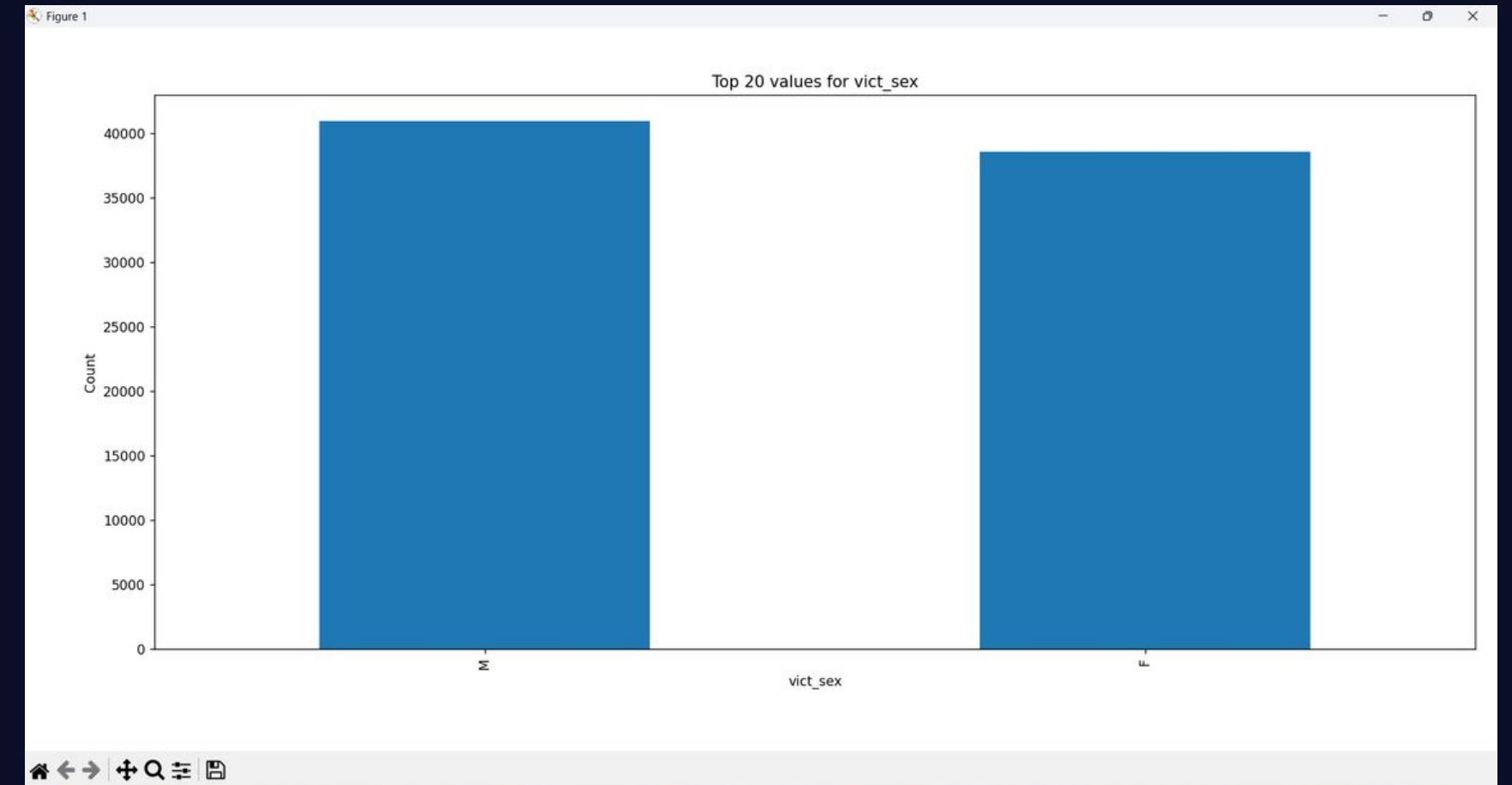
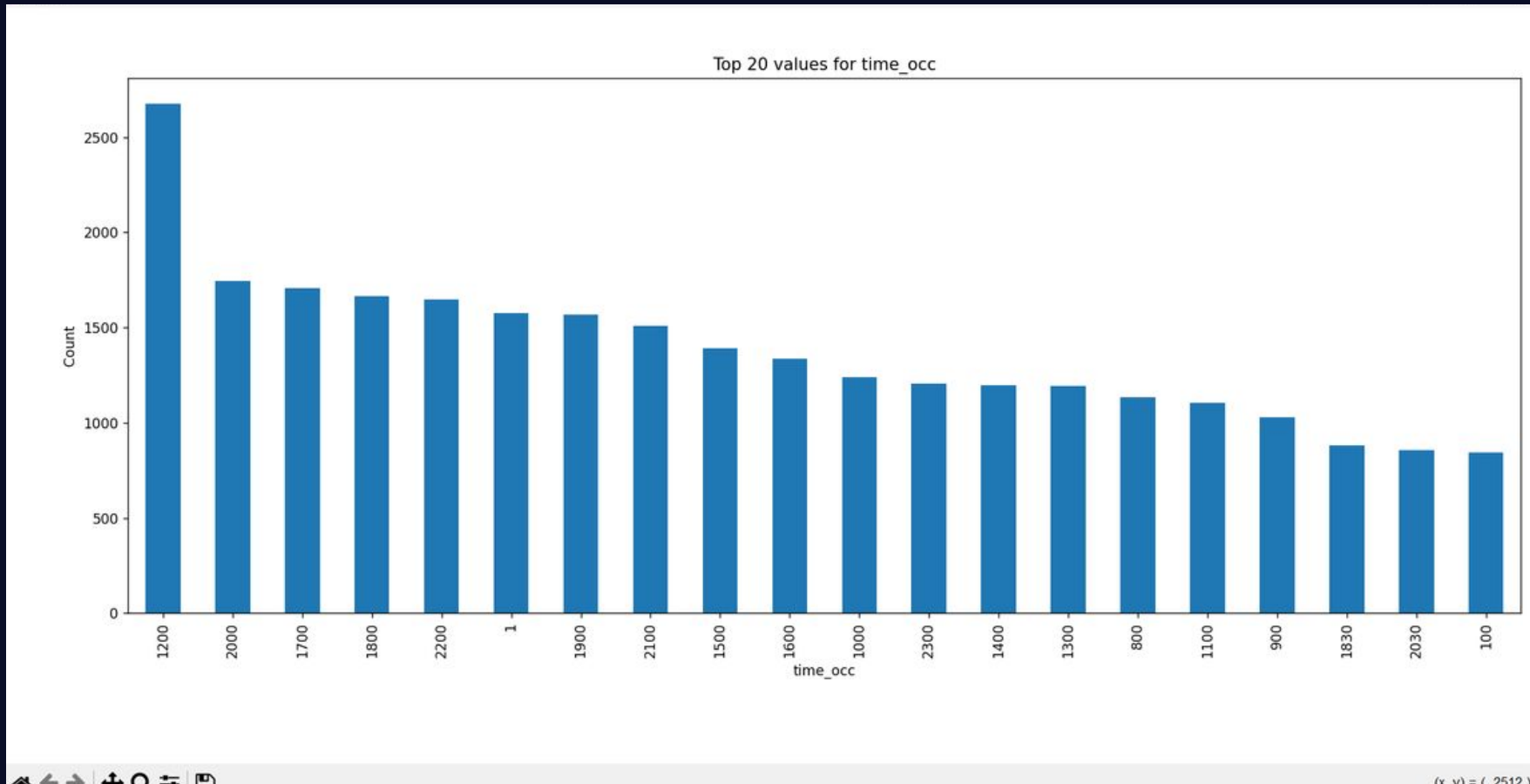
- Checked for missing values (0, NAs)
- Removed or standardized column names and removed invalid strings
- Fixed date formats for DATE_RPTD and DATE_OCC
- Converted numeric fields and cleaned invalid ages
- Dropped high-missing-value columns
- Enforced valid LA coordinate ranges
- Removed rows missing essential information

Data Transformation

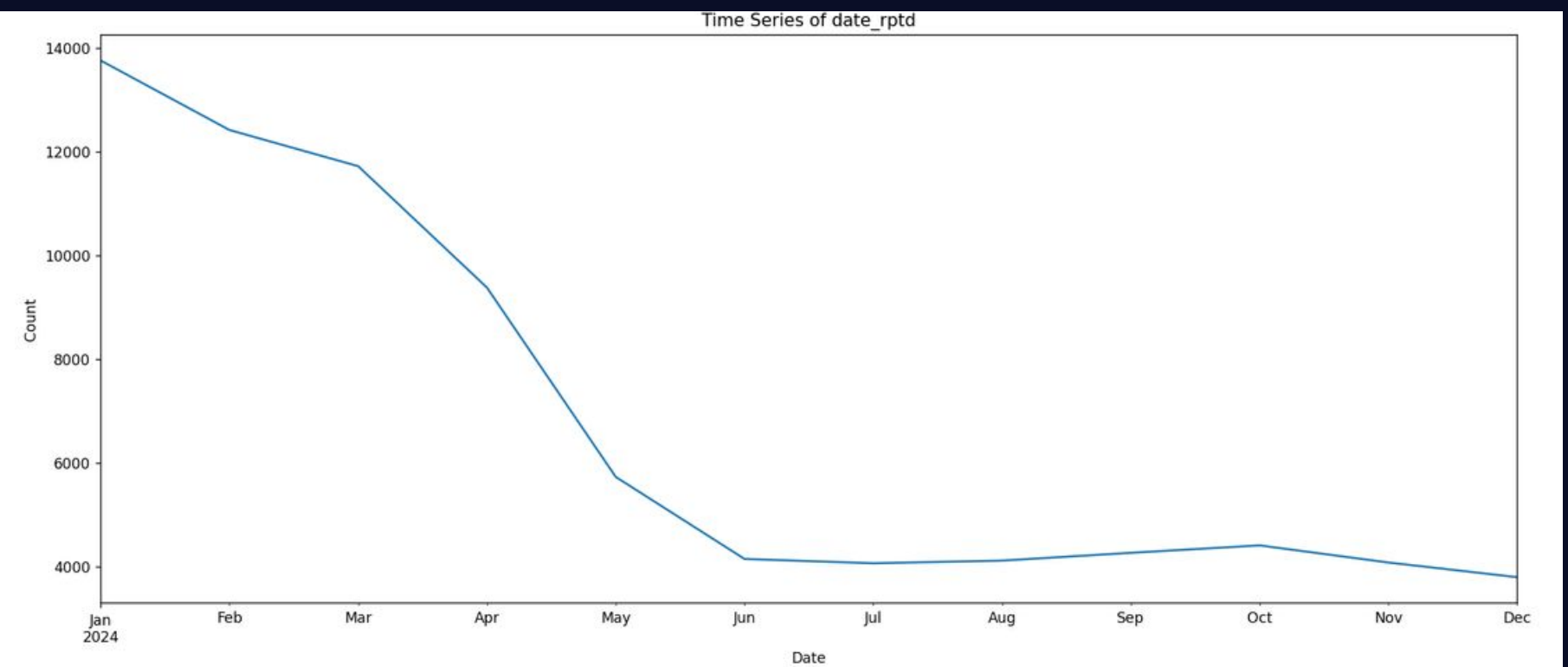
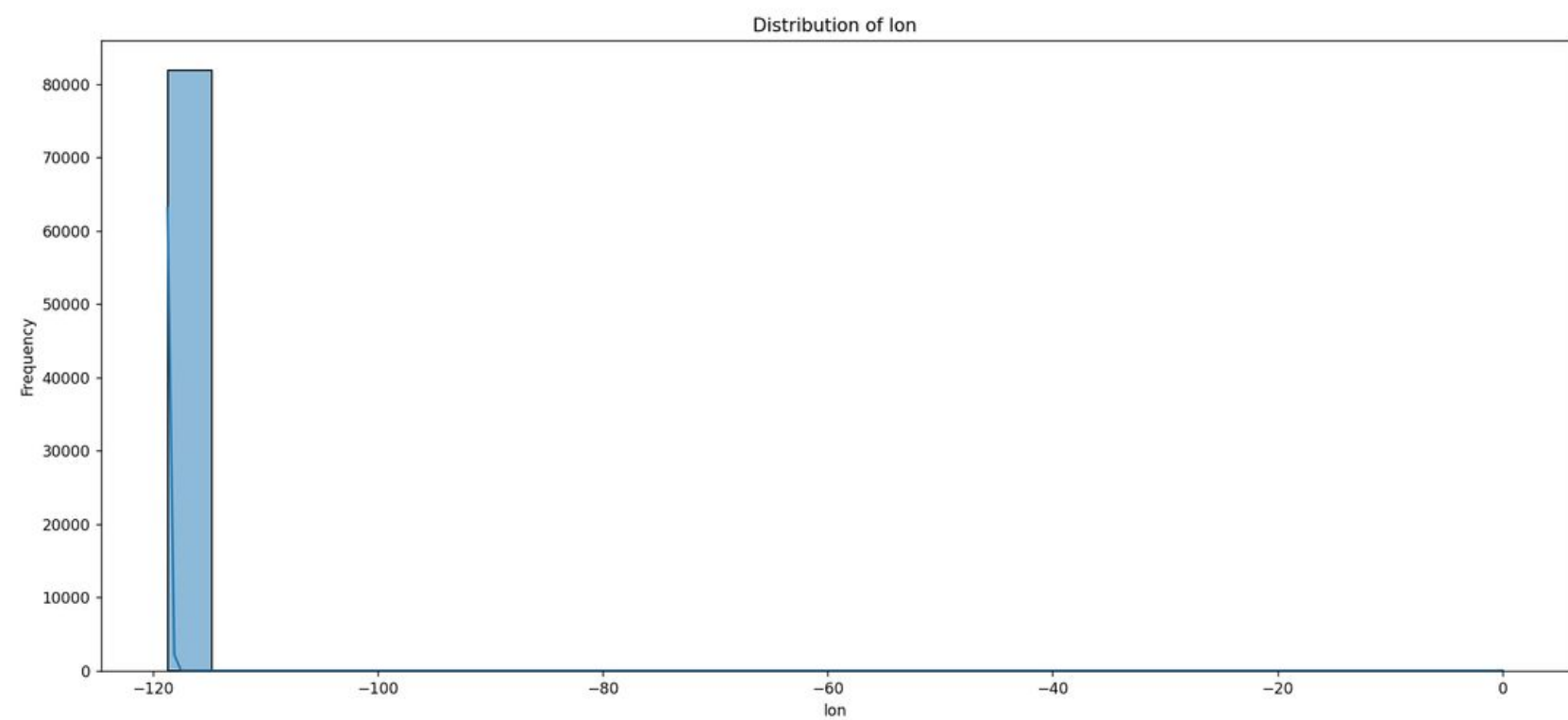
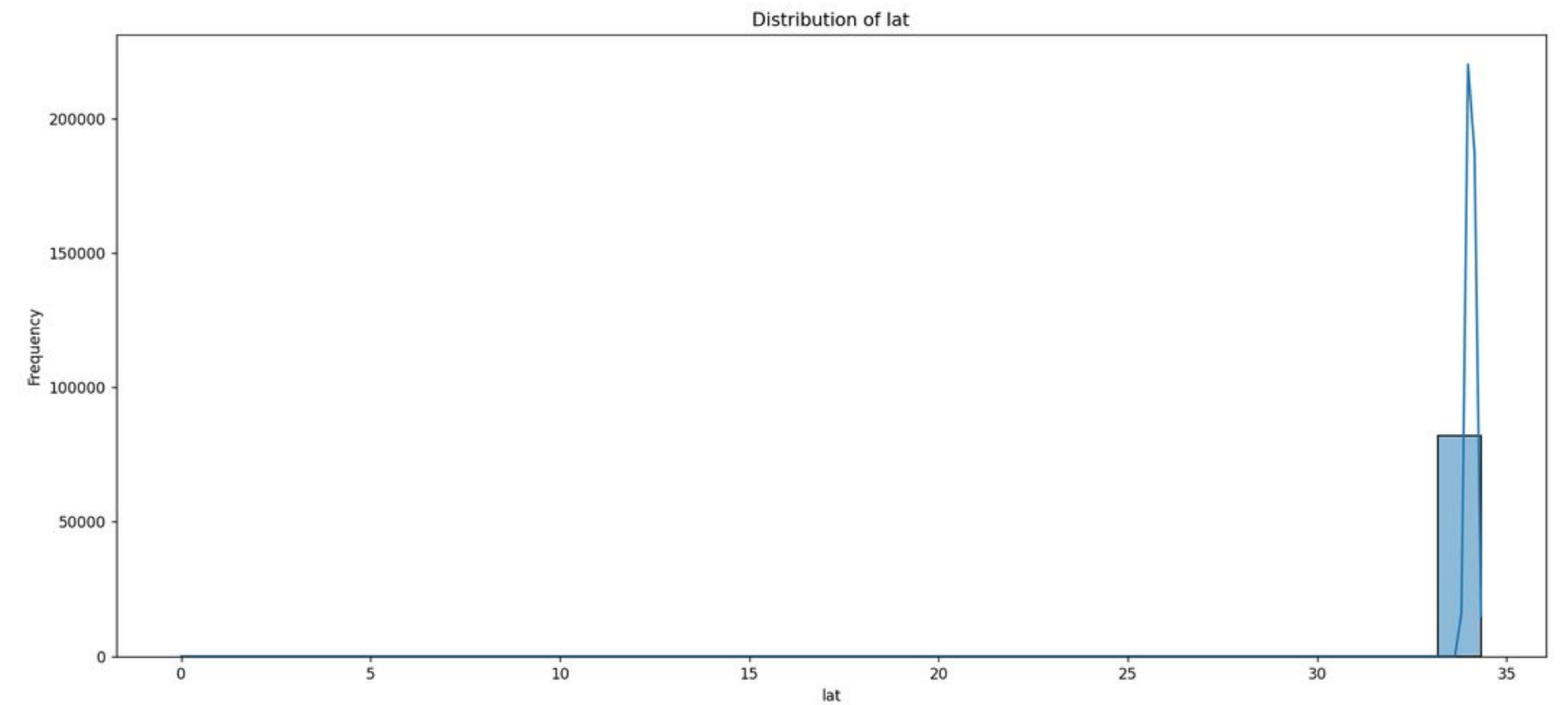
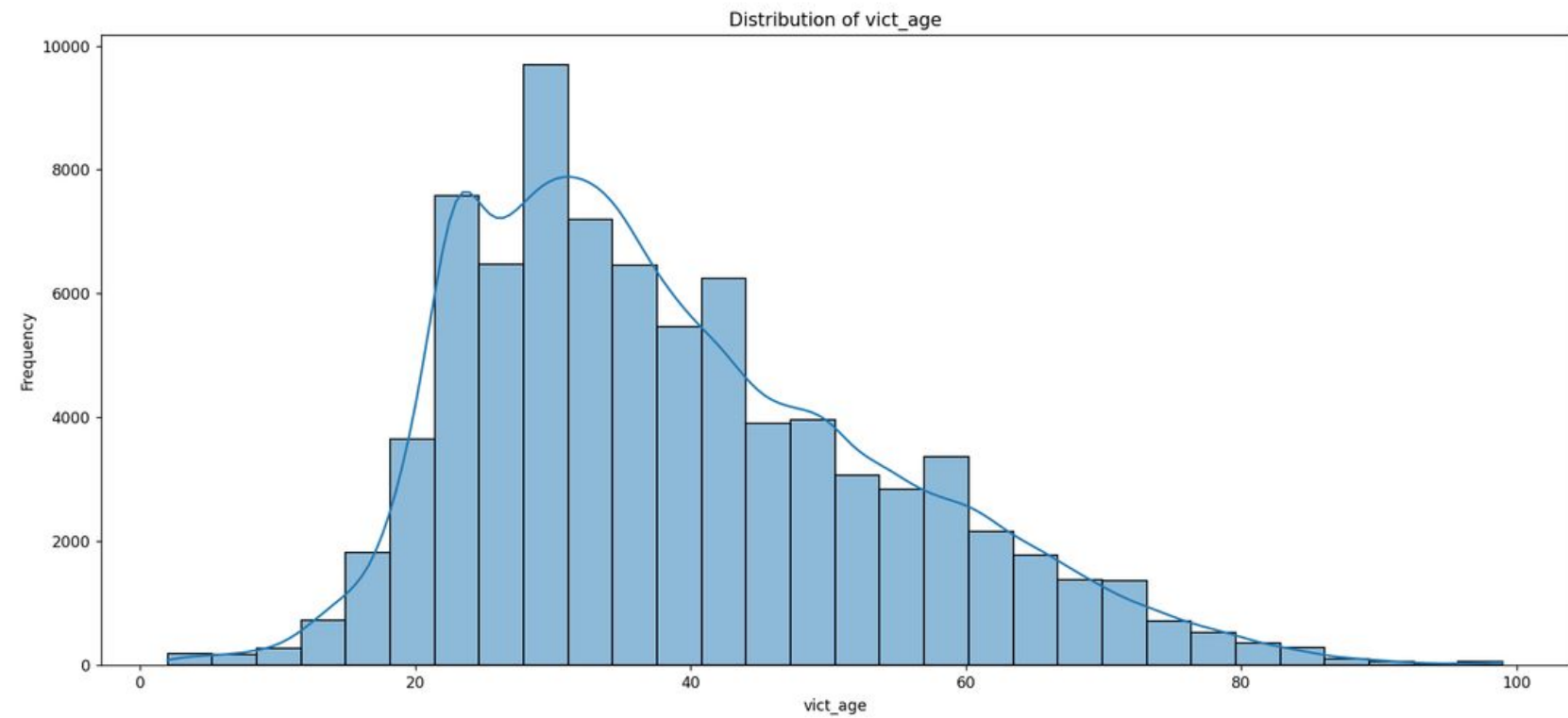
Transformations Applied:

- Extracted month from date_occ for trend analysis
- Aggregated crime counts by month and crime type
- Built pivot tables for start-to-end trend comparison
- Created crime-intensity dataset for modeling
- One-hot encoded categorical fields
- Used plots in the EDA to check for distribution of data
- Prepared final clean dataset for statistical tests & ML

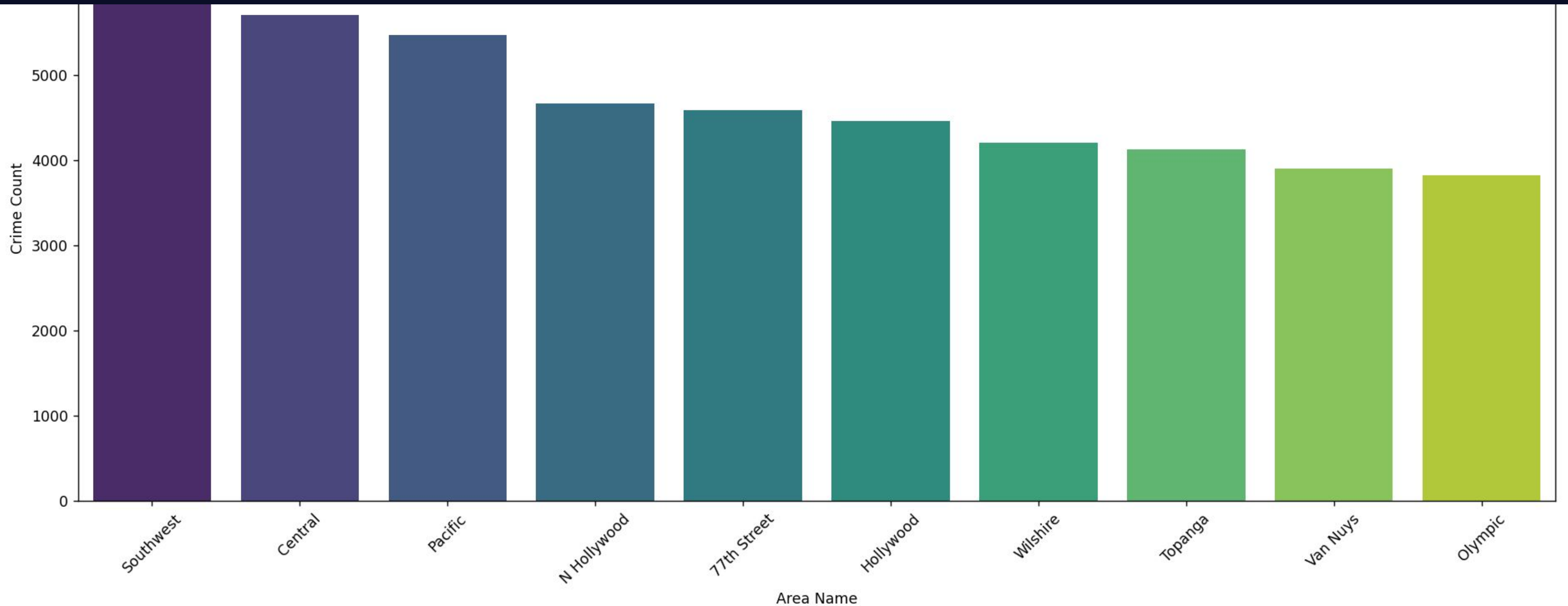
Exploratory Data Analysis



Exploratory Data Analysis



Which areas had the highest crime concentration in 2024?



Top 10 Areas with Highest Crime Counts in 2024

<u>Area Name</u>	<u>Crime Count</u>
------------------	--------------------

Southwest	6561
Central	5709
Pacific	5478
N Hollywood	4668
77th Street	4595
Hollywood	4467
Wilshire	4211
Topanga	4128
Van Nuys	3908
Olympic	3825

Futher Analysis: Statistical Testing and Modeling

a. Confirmatory statistical tests (Are crime rates significantly different across areas on the basis of crime seriousness?)

- We created a contingency table targeting area_name and part 1_2 (representing crime seriousness). 1: serious crime; 2: less serious crime
- And, performed a chi-square test to see if crime seriousness differ across all the areas.

Chi-Squared Test Results:

Chi2 Statistic: 1643.9404471333796

P-Value: 0.0

Degrees of Freedom: 20

Result: Significant difference in crime rates across areas (reject H0).

MODELLIN G

b. What factors explain crime intensity?

- Predicting crime_count using victim & location characteristics
- Aggregating the raw crime dataset into area-level statistics.
- Using a **NEGATIVE BINOMIAL REGRESSION** model to predict crime intensity (crime_count) using demographic and location characteristics.
- Evaluating the model with RMSE and R^2 .
- Showing which variables are most important for predicting crime intensity.

Generalized Linear Model Regression Results						
=====						
Dep. Variable:	crime_count	No. Observations:	21			
Model:	GLM	Df Residuals:	13			
Model Family:	NegativeBinomial	Df Model:	7			
Link Function:	Log	Scale:	1.0000			
Method:	IRLS	Log-Likelihood:	-194.16			
Date:	Thu, 04 Dec 2025	Deviance:	0.48674			
Time:	16:58:59	Pearson chi2:	0.472			
No. Iterations:	6	Pseudo R-squ. (CS):	0.04569			
Covariance Type:	nonrobust					
=====						
	coef	std err	z	P> z	[0.025	0.975]

Intercept	8.2454	0.218	37.780	0.000	7.818	8.673
mean_victim_age	-0.2302	0.597	-0.385	0.700	-1.401	0.940
pct_male_victims	0.1110	0.415	0.267	0.789	-0.703	0.925
pct_female_victims	-0.0281	0.344	-0.082	0.935	-0.702	0.646
mean_lat	-0.0990	0.298	-0.332	0.740	-0.683	0.485
mean_lon	-0.1469	0.482	-0.305	0.760	-1.091	0.797
unique_crime_types	0.0626	0.375	0.167	0.867	-0.672	0.797
unique_locations	0.0959	0.394	0.243	0.808	-0.676	0.868
=====						

RATE RATIOS (INTERPRETABLE EFFECTS)

		Predictor	Coefficient	RateRatio(exp(coef))
Intercept		Intercept	8.245365	3809.926694
mean_victim_age	mean_victim_age		-0.230158	0.794408
pct_male_victims	pct_male_victims		0.110986	1.117379
pct_female_victims	pct_female_victims		-0.028098	0.972293
mean_lat	mean_lat		-0.098981	0.905760
mean_lon	mean_lon		-0.146878	0.863399
unique_crime_types	unique_crime_types		0.062570	1.064569
unique_locations	unique_locations		0.095910	1.100660

McFadden Pseudo-R²: 0.669

- Areas with more unique locations → tend to have more crime
- Areas with more unique crime types → tend to have more crime
- Areas with younger victims → tend to have more crime

The Negative Binomial regression produced a good overall model fit (McFadden pseudo-R² = 0.669), indicating that the area-level predictors collectively explain a substantial portion of the variation in crime intensity.

Do crime incidents differ between weekdays and weekends in Los Angeles?

- Compare crime levels between weekdays and weekends
- Cascading analysis: daily, hourly, time-of-day

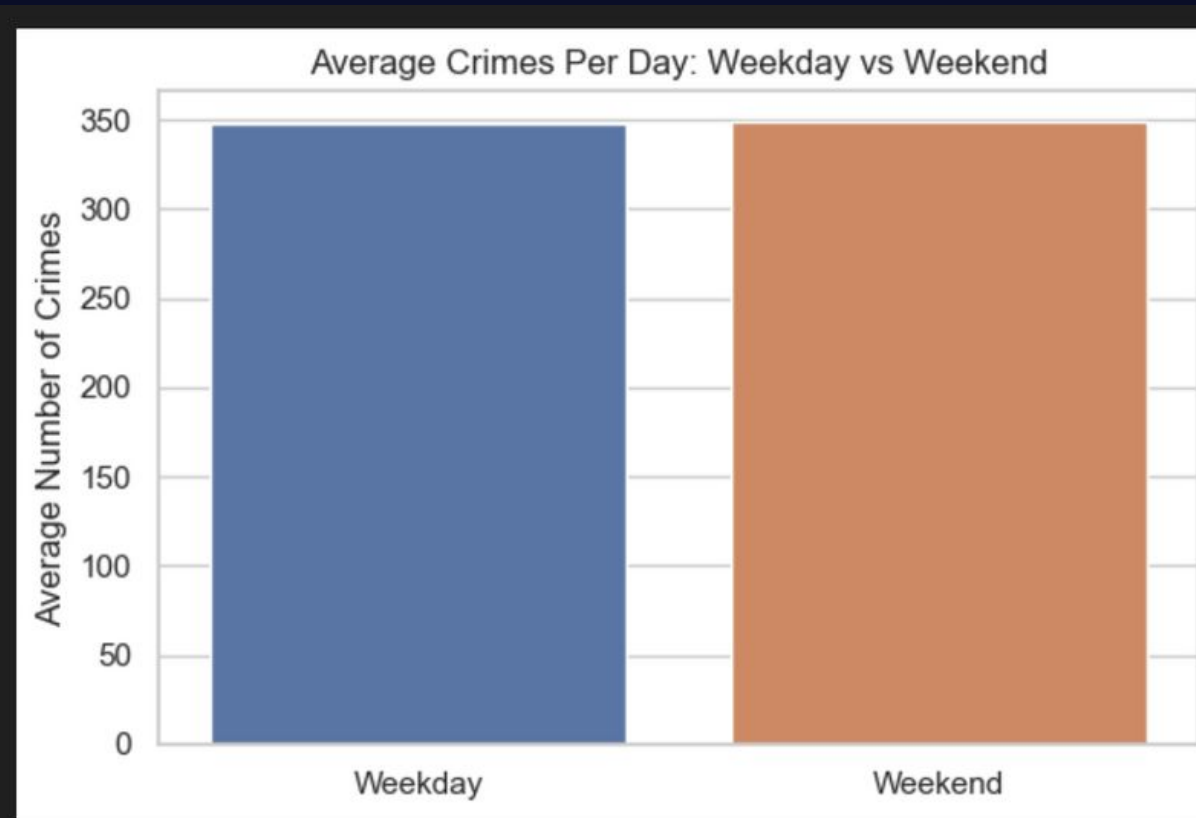
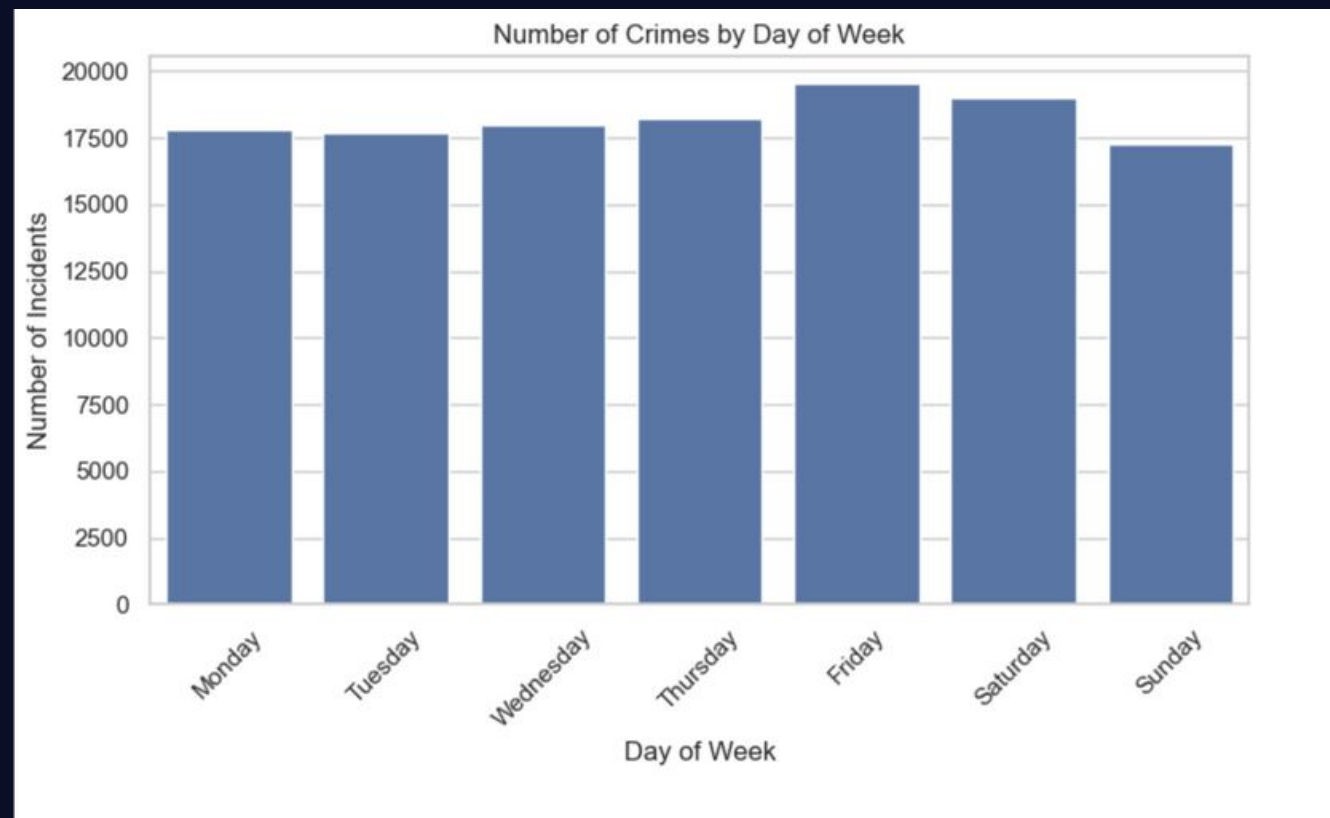


How we answered the Question

- Created features: day of week, weekend flag, hour, time-of-day category
- Compared crimes per day
- Compared crime per hour
- Compared time-of-day groups (Night, Morning, Afternoon, Evening)
- Used statistical tests (Welch t-test, Chi-Square) to validate differences
- **DATE OCC**
- **TIME OCC**
- **DAY_OF_WEEK**
- **IS_WEEKEND**
- **HOUR**
- **TIME_OF_DAY**



Crimes by Day of Week



Hypotheses:

H0: Mean daily crime is the same on weekdays and weekends.

H1: Mean daily crime is different.

Result:

- t-statistic ≈ 0.0678
- p-value ≈ 0.9460 (very large)

There is no statistically significant difference in mean daily crime levels

between weekdays and weekends

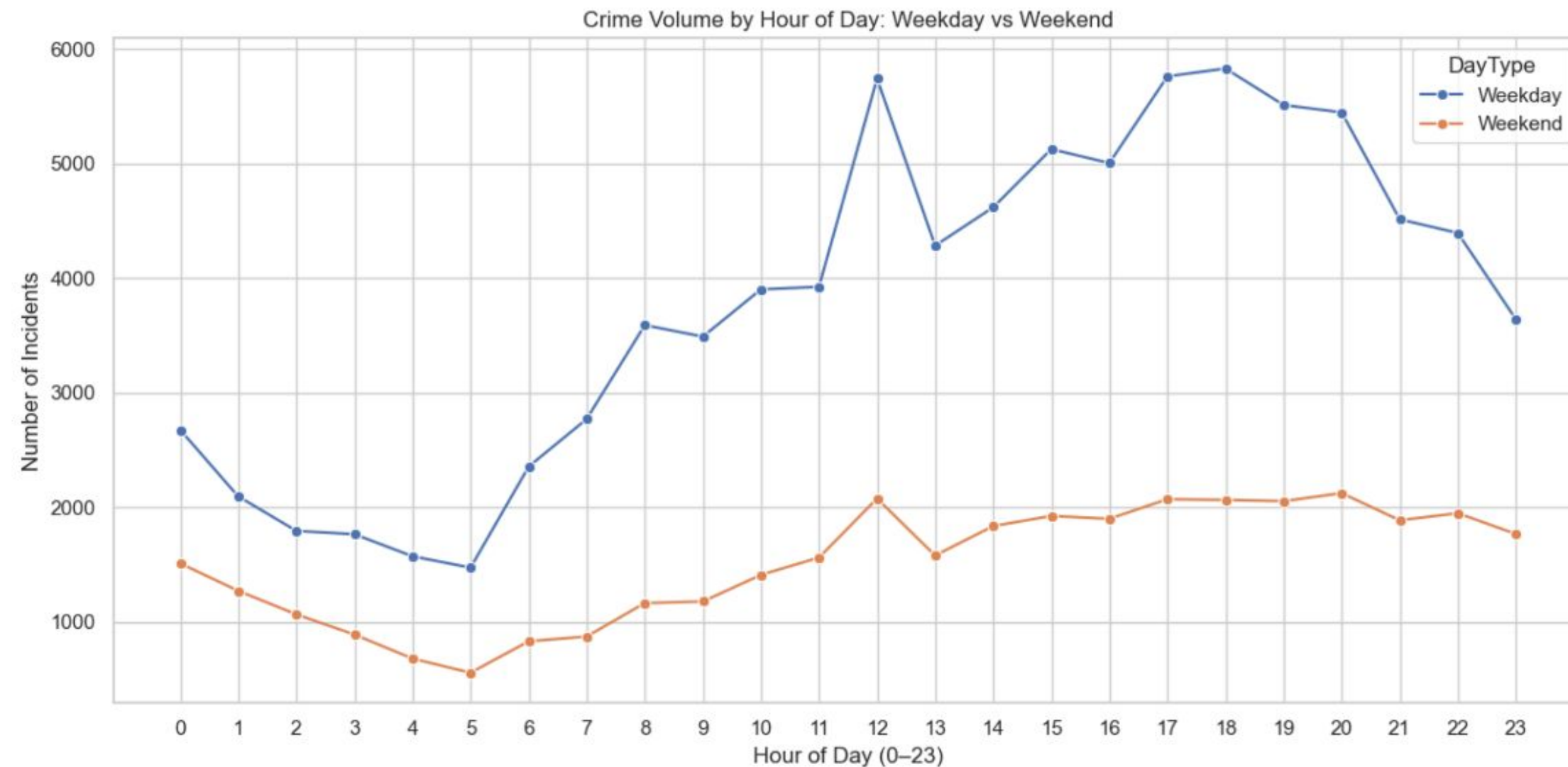
Since daily crime levels do not show a meaningful difference, we move one level

deeper and examine whether crimes differ at the hourly level instead of

just daily totals



Hourly Crime



Hypotheses:

H0: The mean number of crimes per hour is the same on weekdays and weekends.

H1: The mean number of crimes per hour is different.

Result:

- t-statistic ≈ 0.346
- p-value ≈ 0.729

- There is no statistically significant difference in mean daily crime levels between weekdays and weekends
- Both daily and hourly average tests show no significant difference between weekday and weekend crime levels.
- We now move to a categorical time-of-day comparison (Night / Morning / Afternoon / Evening), using a Chi-square test on distributions instead of means.



Time-of-Day Categories

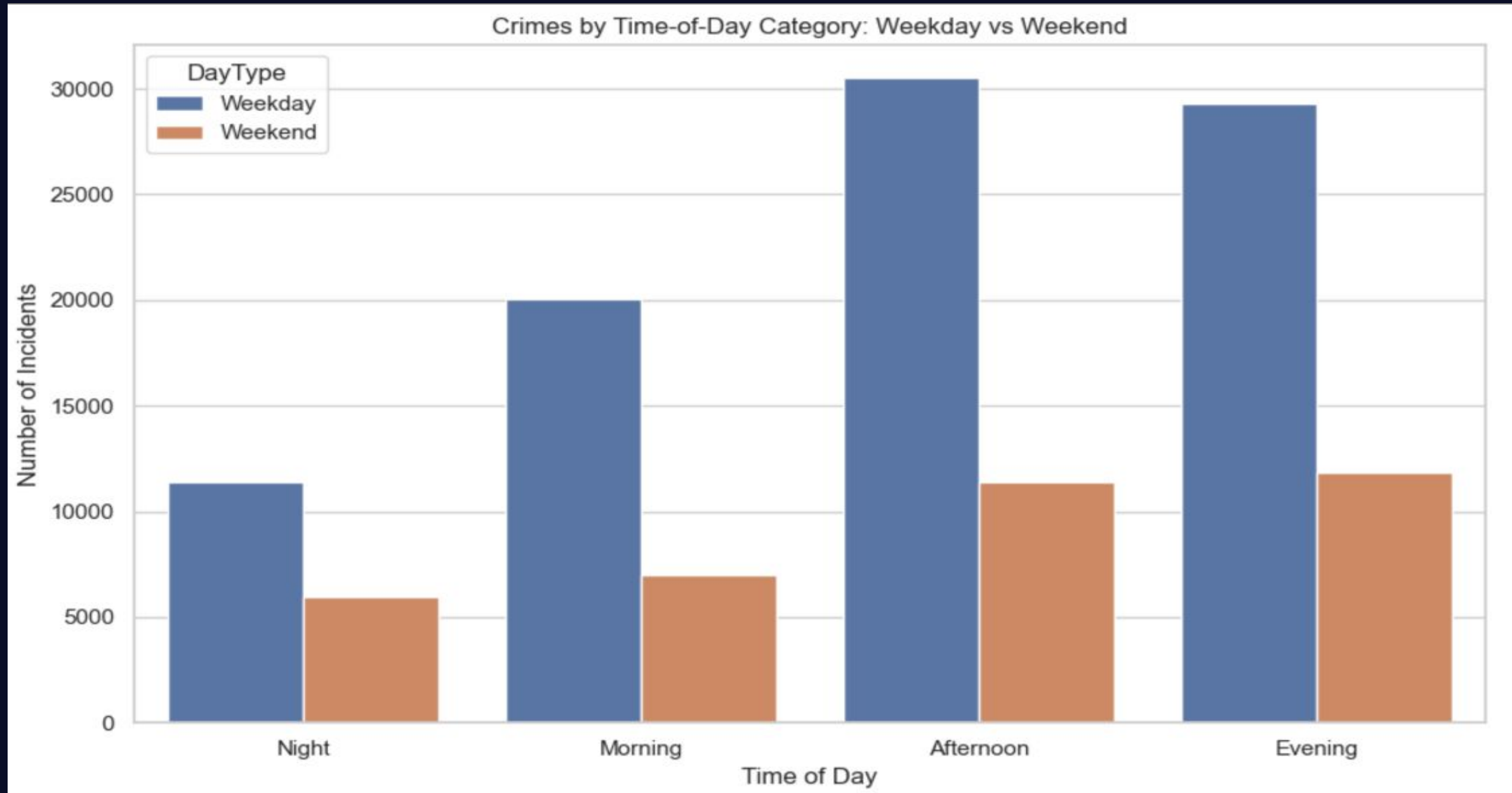
Hypotheses:

H0: Crime distribution across Time-of-Day categories

(Night, Morning, Afternoon, Evening) is the same for weekdays and weekends.
H1: The distribution is different between weekdays and weekends.

Result:

- Chi-square statistic ≈ 424.32
- Degrees of freedom = 3
- p-value $\approx 1.19e-91$ (extremely small)

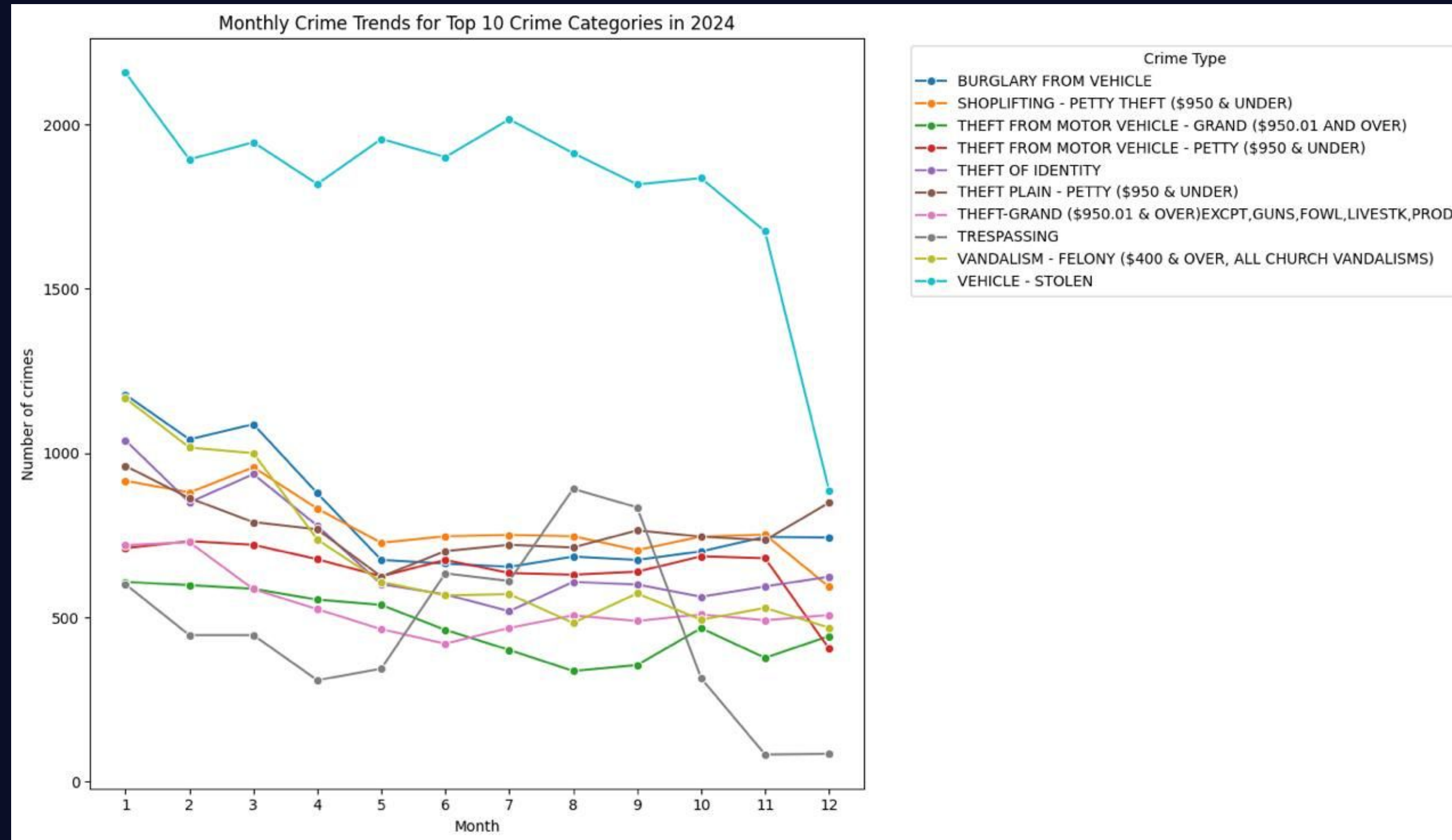


There is a highly significant difference in how crimes are distributed across time-of-day categories on weekdays versus weekends. This means that even though the total daily and average hourly crime levels are similar, the timing of when crime happens throughout the day shifts meaningfully between weekdays and weekends.



Which crime types showed the sharpest increases or decreases in 2024?

- Most major crime categories decrease over the year, especially theft-related crimes
- Only a few categories show short-term increases, but none display an overall upward trend in this top-10 group
- The graph visually confirms a city-wide downward trend in major crimes across 2024



Models Summary

- Crime trends in 2024 were not uniform — some categories rose while many fell.
- The declines affect more severe crimes (homicide, vandalism, shots fired).
- The increases affect opportunistic thefts (pickpocketing, boat theft).
- Statistical tests confirm the trends are significant and reliable, not noise

===== SIGNIFICANT INCREASING TRENDS (Kendall's Tau) =====					
	crime_type	tau	p_value	trend_direction	significant
9	BOAT – STOLEN	0.676123	0.030464	Increasing	YES
64	PICKPOCKET	0.515152	0.020980	Increasing	YES
===== SIGNIFICANT DECREASING TRENDS (Kendall's Tau) =====					
	crime_type	tau	p_value	\	
33	CRUELTY TO ANIMALS	−0.948683	0.022977		
10	BOMB SCARE	−0.883883	0.001326		
36	DISCHARGE FIREARMS/SHOTS FIRED	−0.816982	0.002366		
97	VANDALISM – MISDEAMEANOR (\$399 OR UNDER)	−0.809184	0.000270		
23	CHILD PORNOGRAPHY	−0.808224	0.003662		
5	BATTERY ON A FIREFIGHTER	−0.793884	0.017235		
96	VANDALISM – FELONY (\$400 & OVER, ALL CHURCH VA...	−0.787879	0.000107		
30	CRIMINAL HOMICIDE	−0.772252	0.000660		
19	CHILD ABUSE (PHYSICAL) – AGGRAVATED ASSAULT	−0.750939	0.021143		
47	FALSE POLICE REPORT	−0.748355	0.007132		

Conclusion :

Our analysis of Los Angeles crime data for 2024 shows clear geographic clustering, with Southwest, Central, and Pacific emerging as the highest-crime areas. While overall crime volumes decreased for major categories such as burglary, vehicle theft, and simple assault, a few offenses including pickpocketing and boat theft showed meaningful increases confirmed through Kendall's Tau significance testing. We also found that total crime levels do not differ significantly between weekdays and weekends; however, the timing of incidents shifts sharply, with a highly significant Chi-square test revealing different time-of-day patterns across the week. Taken together, these findings show that crime in Los Angeles is shaped by strong spatial patterns, evolving monthly trends, and meaningful behavioral differences in when crimes occur, highlighting the importance of data-driven strategies for prevention and resource allocation.

Thank You